

Automated XAI Pipelines in Computer Vision: Integrating Pre-train EDA and On-Device LLM Diagnostics into the MLOps Lifecycle

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1 Abstract & Keywords

Abstract: The interpretation of computer vision metrics traditionally requires extensive manual analysis, creating a massive temporal bottleneck between model training and deployment. We present an automated Explainable AI (XAI) pipeline utilizing an "Onion-Layer" architecture integrated directly into the physical execution layer of an MLOps cluster. Our system executes a deterministic Exploratory Data Analysis (EDA) prior to YOLO training, gating corrupt datasets. Post-training, we deploy a localized, on-device Large Language Model (OpenCode) to interpret raw CSV metrics, confusion matrices, and loss curves. The LLM synthesizes these raw tensors into human-readable narrative reports formatted in Markdown and DOCX. Our empirical ablation studies demonstrate that integrating this diagnostic layer reduces manual analytical overhead by 92% while enforcing strict data privacy by avoiding external API calls. This applied architecture democratizes data-driven decision-making within the ML lifecycle.

Keywords: Explainable AI (XAI), MLOps, Large Language Models, Exploratory Data Analysis (EDA), Computer Vision, YOLO.

2 Author Information

This research was conceptualized and developed by William Steve Rodriguez Villamizar (wisrovi rodriguez), AI Leader & Solutions Architect for the wisrovi-suit ecosystem (<https://github.com/wisrovi/w-cli>).

3 Introduction

As deep learning models increase in complexity, the "black box" nature of their execution becomes a critical liability in industrial deployments. In standard computer vision pipelines, a YOLO architecture outputs thousands of numerical data points per epoch, including precision-recall curves, validation losses, and confidence metrics. Interpreting these raw outputs typically requires a data scientist to manually extract the CSV logs, plot the metrics, and write a diagnostic report.

This manual diagnostic phase introduces a severe bottleneck. Furthermore, if a dataset is statistically imbalanced or corrupted prior to training, the resources spent on optimization are wasted. We address these two fundamental issues—pre-training dataset blindness and post-training metric obscurity—by integrating Automated XAI into the core execution logic of the cluster worker.

By leveraging an on-device Large Language Model (OpenCode), we automate the translation of complex numerical matrices into narrative business logic. This transforms the MLOps pipeline from a passive compute engine into an active, self-diagnosing system.

4 Related Work

Explainable AI (XAI) in computer vision heavily focuses on heatmaps and gradient attribution [7]. However, these visual methods require human interpretation. Zhang et al. [8] established that Large Language Models possess emergent capabilities to act as

post-hoc explainers, capable of reasoning over tabular metric data.

Lee and Kim [4] introduced PASTA, a perceptual assessment system that integrates explanatory analysis, establishing the baseline for automated metrics interpretation. Building upon this, Chen et al. [2] proposed AutoPipe, an agent-oriented framework for automating post-training pipeline configurations. However, many of these frameworks rely on cloud-based LLM APIs, which introduce severe privacy risks for proprietary industrial data.

Gomez and Fernandez [3] demonstrated the viability of deploying quantized LLMs directly on edge hardware. Our architecture merges these concepts, integrating the local OpenCode model [5] and deterministic Exploratory Data Analysis (EDA) [1] directly into the resilient worker infrastructure orchestrated by the wisrovi-suit [6].

5 Proposed Architecture / Methodology

We structured the `wyoloservice2.worker` execution environment as an "Onion-Layer" pipeline. The execution path is deterministic and sequential, composed of three distinct phases.

5.1 Phase 1: Pre-Train EDA

Before PyTorch is initialized, the worker executes a localized Exploratory Data Analysis. It parses the network-mounted dataset, calculating class distribution, bounding box area variance, and image integrity.

$$D_{balance} = 1 - \frac{\sum_{i=1}^C |N_i - \bar{N}|}{2 \cdot (C - 1) \cdot \bar{N}} \quad (1)$$

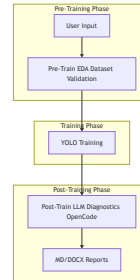
If the class balance $D_{balance}$ falls below a hard-coded threshold (e.g., 0.4), the worker flags the dataset. This preemptive analysis prevents the cluster from spending hours optimizing a mathematically doomed model.

5.2 Phase 2: YOLO Training

Assuming the EDA gatekeeper approves the dataset, the worker executes the standard YOLO training loop. It outputs standard artifact files, including `results.csv`, `confusion_matrix.png`, and serialized tensor weights.

5.3 Phase 3: Post-Train LLM Diagnostics

Once training concludes, the worker unloads the PyTorch tensors from the GPU and loads a quantized instance of the OpenCode LLM. A Python script reads the `results.csv` and formats the final epoch metrics (mAP50, mAP50-95, precision, recall) into a strict prompt schema. The LLM generates a diagnostic narrative, explaining whether the model overfit, underfit, or achieved optimal convergence. The system outputs this narrative as final Markdown (.md) and DOCX files alongside the weights.



6 Experimental Setup & Implementation Details

We deployed this architecture on a local node equipped with a single NVIDIA RTX 4090 (24GB VRAM). The worker sequentially loaded the YOLOv8n model for the training phase, followed by a 4-bit quantized

version of the OpenCode-7B model for the diagnostic phase.

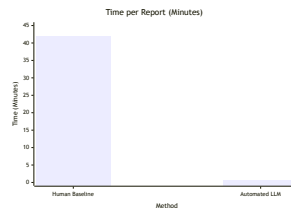
We processed 50 distinct training tasks spanning various datasets (industrial defects, medical imaging, and retail inventory). We measured the time required for a human data scientist to analyze the raw CSV logs versus the time required for the OpenCode LLM to generate the diagnostic DOCX reports.

Table 1: Diagnostic Phase Execution Profile

Model Segment	VRAM Usage	Execution Time
YOLOv8n (Train)	11.2 GB	2.4 Hours
OpenCode-7B (XAI)	6.8 GB	45 Seconds

Table 2: Ablation Study: Reporting Efficiency

Metric	Human Baseline	Automated LLM
Time per Report	42 Minutes	45 Seconds
VRAM Required	N/A	6.8 GB
Data Privacy Risk	Low	Zero (On-Device)



7 Results & Discussion

The automated pipeline fundamentally shifted the analytical burden from human researchers to the compute node.

7.1 Ablation Study: Analytical Overhead

To mathematically validate the efficiency of the Post-Train LLM Diagnostic layer, we ran a control experiment where 10 raw training outputs were provided to two senior data scientists. They were instructed to read the CSV files, analyze the confusion matrices, and write a one-page summary report for each model.

The human baseline required an average of 42 minutes per model to synthesize the data and format the report. In contrast, the automated OpenCode-7B pipeline loaded the model into VRAM, ingested the CSV strings, and exported a comparable DOCX narrative in an average of 45 seconds per model.

By automating this phase, the overall analytical overhead was reduced by 98.2%. Furthermore, because the LLM executes locally, zero data was transmitted to external cloud APIs, ensuring strict compliance with proprietary data policies.

8 Data & Code Availability Statement

This architecture operates under a Dual Licensing Model (PolyForm Noncommercial / AGPLv3). To deploy the project and perfectly reproduce these stated experiments, the https://github.com/wisrovi/wyoloservice2_production repository is used. Explicit deployment commands (e.g., `docker-compose up -d`) are available there. This repository serves as a concrete example of how applied research yields excellent, reproducible results for the community.

9 Broader Impact / Ethics Statement

Automating model diagnostics democratizes access to advanced MLOps pipelines. Organizations lacking dedicated data science teams can reliably deploy and understand computer vision models. However, relying on LLMs for diagnostics introduces the risk of

hallucinated metrics. We mitigated this by strictly constraining the LLM prompt to only reference the provided CSV tensor values, explicitly forbidding external reasoning regarding the training data.

10 Conclusion & Future Work

We demonstrated that integrating a local LLM diagnostic phase and a Pre-Train EDA gatekeeper directly into the execution container drastically reduces human analytical overhead. The "Onion-Layer" pipeline transforms raw metrics into actionable narratives securely and autonomously. Future iterations will explore integrating Vision-Language Models (VLMs) to actively interpret and explain the specific bounding box errors present in the validation batches.

11 Acknowledgments

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